



ALPHA CHOICE INNOVATIVE ACADEMY
(International Secondary School)

4/6 Richard Okoroike Close, Praise Hill Estate, Arepo, Ogun State.

GRADE 8
THIRD TERM
CHEMISTRY
MASTERNOTE

NAME: _____

CHEMISTRY SYNOPSIS FOR GRADE 8

WEEK

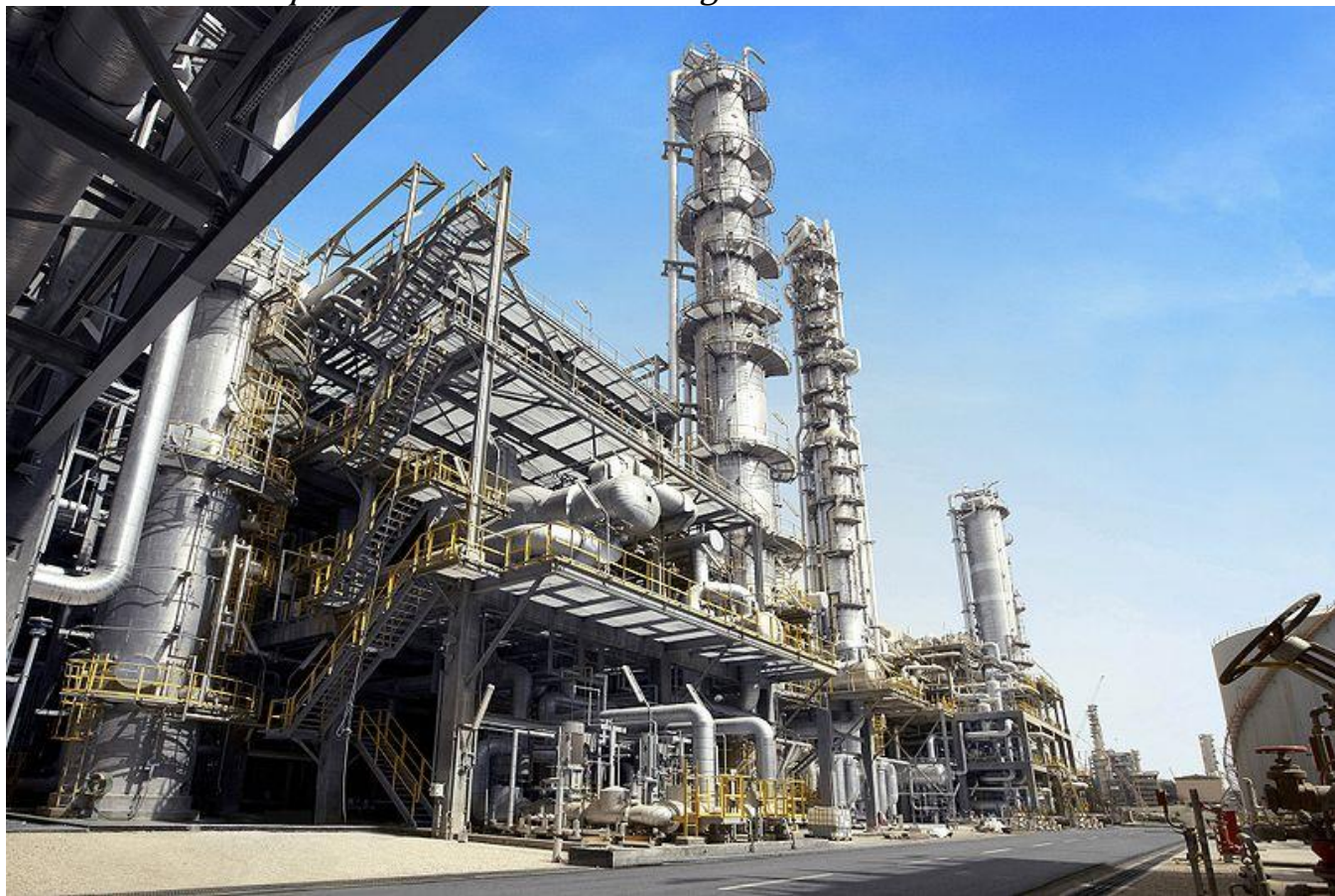
SKILL

1. Crude oil and petrol chemicals
2. Crude oil and petrol chemicals (2)
3. Chemical Symbols and Formula
4. Empirical Formula and Molecular Formula
5. Laws of Chemical Combination
6. Chemical Equations
- 7. Midterm break***
8. Chemical reactions
9. Types of chemical reactions I
10. Types of chemical reactions II

WEEKS 1 AND 2:CRUDE OIL AND PETROCHEMICALS

OBJECTIVES :

- *Explain what crude oil and petrochemicals are.*
- *Describe the process of refining crude oil.*
- *State the uses of crude oil and petrochemicals.*
- *State the importance of crude oil to Nigeria*



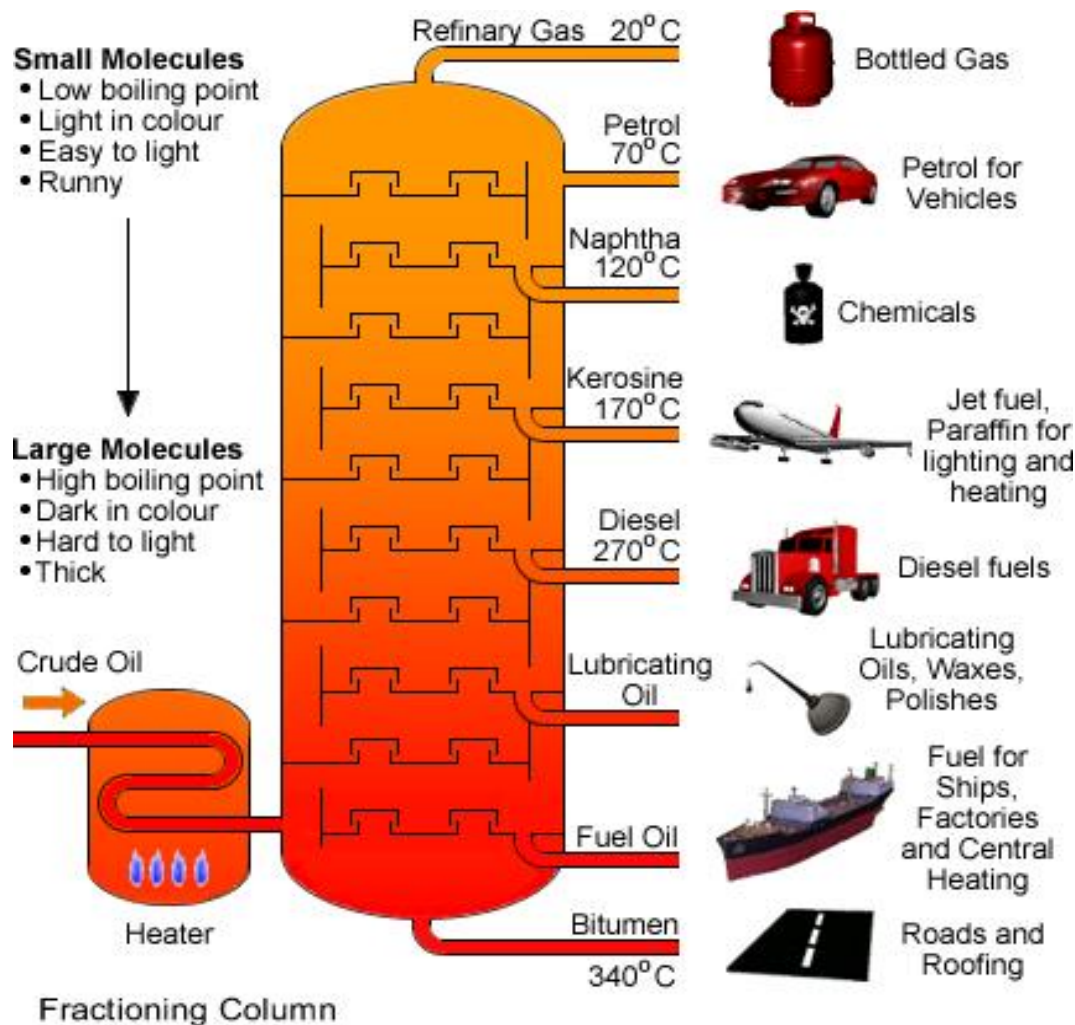
Crude oil or petroleum is a dark, viscous liquid found in huge underground deposits in many parts of the world.

It is formed in the rocks of the earth's crust from the remains of marine animals and plants. It is a complex mixture of gases, liquids and solid hydrocarbons together with some impurities of sulphur, nitrogen and oxygen – sulphur being the most troublesome. Nigerian crude oil is relatively sulphur free and this makes it very marketable.

CRUDE OIL REFINING AND TREATMENT OF FRACTIONS

The separation of crude oil into various useful parts is called **refining**, and this is done in a **refinery** through a process called **fractional distillation**. The separation is carried out in tall towers called **fractionating towers/columns**.

The crude oil is heated in an electric furnace to a high temperature before being conducted in pipes to the fractionating column. In the fractionating column, the vapour of the different fractions/components of the crude oil rise at different speed and to different heights since they have different weights. The fraction with lower boiling points gets to the top while those with lower boiling points are obtained at the lower parts.



<i>Fraction</i>	<i>Boiling Range (°C)</i>	<i>State of matter</i>	<i>Number of Carbon Atoms</i>	<i>Uses</i>
Petroleum gas	< 20	Gaseous	C ₁ to C ₄	Fuel for homes and industries, for producing hydrogen, ethyne, Carbon (IV) sulphide, tetrachloromethane

Petroleum ether & ligroin	20 – 60	Very volatile liquid	C ₅ to C ₇	Organic solvent making
Gasoline or Petrol	40 – 200	Volatile liquid	C ₄ to C ₁₂ , but mostly C ₆ to C ₈	Fuel for aeroplanes, automobile, as solvents for paints and grease.
Kerosene	200 – 250	Fairly volatile liquid	mostly C ₁₂ to C ₁₈	Fuel for lighting and heating, driving tractors & some modern jet engines, good solvent for grease and paints
Fuel oils - gas oil & diesel oil	260 – 350	Liquid	C ₁₂ to C ₂₅	Fuel for heating and diesel engines, raw material for cracking process.
Lubricants or lubricating oil	350 – 500	Viscous liquid/solid	C ₂₀ and above	Lubricating moving parts of engines, Vaseline – solid for hair care product Paraffin wax – solid for candle, water proof, polish, ointment, cream
Asphalt Bitumen etc	>500	Solid residue		Bitumen for surfacing roads & airfields, others for pipecoating, waterproof roofs, fuel and protective paints

Straight-run gasoline burns unevenly in high-compression engines, producing a shock wave that causes the engine to "knock," or "ping."

An **antiknock agent** is a gasoline additive used to reduce engine knocking and increase the fuel's octane rating. Lead (II) tetra ethyl (Tetra-ethyl lead) is in common use.

Octane Number: Octane number is a value used to indicate the resistance of a motor fuel to knock. Octane numbers are based on a scale from 0 to 100.

A long, straight-chain alkane, which burns unevenly and produces a great deal of knocking has octane number of 0. Highly branched alkanes are more resistant to knocking have octane number of 100.

Further treatment of petroleum products

1. Cracking: Cracking is the breaking down of large hydrocarbon molecules (long-chained) into smaller hydrocarbons (short-chained) and more useful bits of fragments for use in gasoline.

This is achieved by using high pressures and temperatures without a catalyst, or lower temperatures and pressures in the presence of a catalyst.

Due to the high demand for petrol and other useful volatile crude oil fractions, the less volatile fractions such as diesel oil is converted to petrol by cracking.

(a) Thermal cracking: In thermal cracking, high temperatures (about 450°C to 750°C) and pressures (up to about 70 atmospheres) are used to break the large hydrocarbons into smaller ones. Thermal cracking gives mixtures of products containing high proportions of hydrocarbons with double bonds - alkenes.

(b) Catalytic cracking: Involves heating long chain hydrocarbons under pressure and in the presence of catalyst. The alkane is brought into contact with the catalyst at a temperature of about 500°C and moderately low pressures.

Modern cracking uses *zeolites* as the catalyst. These are complex aluminosilicates, containing aluminium, silicon and oxygen atoms. You may have come across a zeolite if you know about ion exchange resins used in water softeners.

NOTE: Catalytic cracking produces petrol mixture rich in branch chain hydrocarbon with an attendant increase in octane number more than thermal cracking.

Cracking is a breaking down process while reforming is an isomerization process.

2. Reforming: This is the process of rearranging the structure of petroleum products (e.g. petrol) to improve its anti-knock properties.

It is a chemical process used to convert petroleum refinery naphtha, typically having low octane ratings, into high-octane liquid products called **reformates** which are components of high-octane gasoline (also known as high-octane petrol).

Reforming is used to improve the quality of petrol by converting straight-chain hydrocarbons to branch-chain and cyclic hydrocarbons which yield high performance fuels. Hydrogen is produced as a by-product of the reactions.

Types of reforming

(a) Thermal reforming.

(b) Catalytic reforming:

While thermal reforming could produce reformat with octane numbers of 65-80 depending on the yield, catalytic reforming produces reformat with octane numbers on the order of 90-95.

Reforming and gasoline additives such as tetraethyl lead increase the octane number of the straight-run gasoline obtained from the fractional distillation of crude oil, but neither process increases the yield of gasoline from a barrel of oil.

PETROCHEMICALS

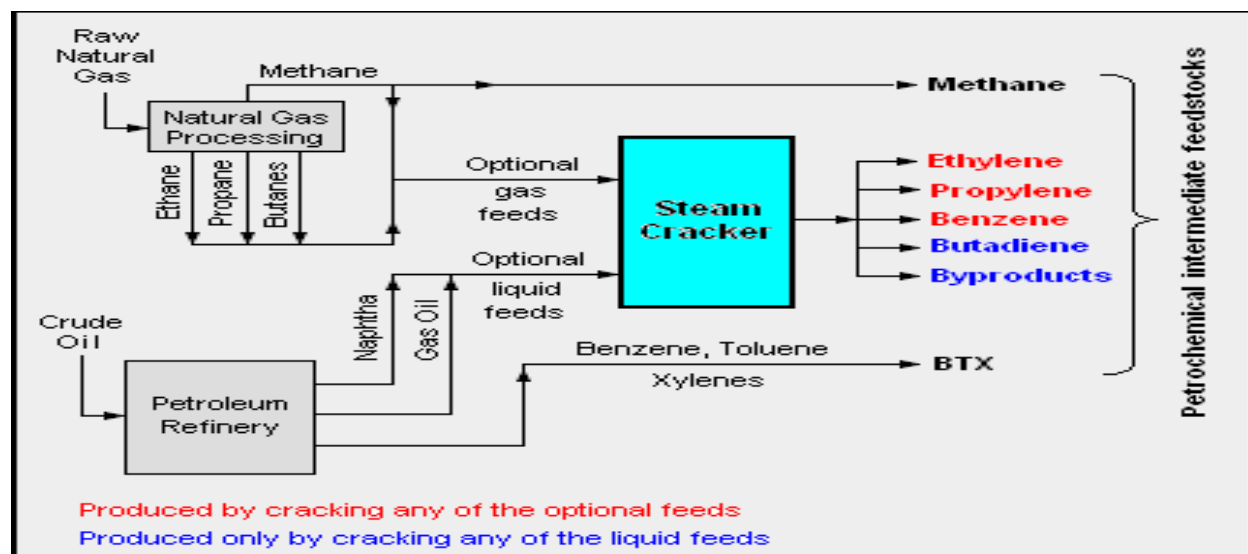
Petrochemicals are substances that are manufactured from the by-products of petroleum. These include plastics, nylon, Dacron, synthetic rubber, insecticides, inorganic fertilizers, benzene, toluene, synthesis gas and detergents

Some chemical compounds made from petroleum are also obtained from other fossil fuels, such as coal or natural gas, or renewable sources such as corn or sugar cane.

Major End Use Products:

Some typical petrochemical intermediates are:

- Vinyl acetate for paint, paper and textile coatings vinyl chloride for polyvinyl chloride (PVC) for water pipes electrical installations.
- Resin manufacture for zeolites etc.
- Ethylene glycol for polyester - textile fibers
- Styrene which is important in rubber and plastic manufacturing.
- Benzene for nylon acid.



STUDY QUESTIONS

1. What is crude oil?
2. Explain what petrochemicals are.
3. Describe the process of refining crude oil.
4. State the uses of crude oil and petrochemicals.
5. What is the importance of crude oil to Nigeria?
6. Briefly explain the process of formation of crude oil
7. Name seven products/ fractions that can be obtained from crude oil.
8. The process of separation of crude oil is called what?
9. What do you understand by cracking?
10. Define octane number and reforming

WEEK 3: CHEMICAL SYMBOLS AND FORMULAE

OBJECTIVES

- Define chemical symbol.
- Write the symbols of the first twenty elements.

A chemical symbol is the abbreviation for the name of an element. It is a letter or letters representing the name of an element. There are four ways of writing the symbols of elements.

- By using the first letter of the name of the element. The symbol is written in capital letter. Examples are:

Name of element	Symbol
Hydrogen	H
Boron	B
Oxygen	O
Nitrogen	N
Fluorine	F
Phosphorus	P
Sulphur	S

- By using the first two letters of the name of the element. The first letter of the symbol is written as a capital letter, while the second letter is written as a small letter. Examples are:

Name of element	Symbol
Helium	He
Lithium	Li
Beryllium	Be
Neon	Ne
Aluminium	Al
Silicon	Si
Argon	Ar
Calcium	Ca

- By using the first letter and any other letter in the name of the element. The first letter of the symbol is written as a capital letter, while the second letter is written as a small letter. Examples are:

Name of element	Symbol
Magnesium	Mg
Chlorine	Cl
Manganese	Mn
Zinc	Zn
Chromium	Cr
Platinum	Pt
Strontium	Sr

4. By using one or more letters from the Latin name of the element. E.g.

English name of element	Latin name	Symbol
Sodium	Natrium	Na
Iron	Ferrum	Fe
Copper	Cuprum	Cu
Potassium	Kalium	K
Silver	Argentum	Ag
Gold	Aurum	Au
Lead	Plumbum	Pb
Mercury	Hydrargyrum	Hg

CHEMICAL FORMULAE

This is a shorthand representation which shows the kind and number of atoms in one molecule of a compound. It is the representation of an element, a molecule or a compound by using symbols of atoms. It is represented by letters and numerals. The letters represent the symbol of the element, while the numerals represent the number of atoms in the molecule or compound.

IUPAC Nomenclature (name) of inorganic compounds

IUPAC means international union of pure and applied chemistry. It is the body of chemists that agrees on the names of chemical substances in order to have uniform name for a particular substance all over the world.

Naming of binary compounds

A binary compound is a compound that is made up of two substances (elements or radicals).

Note: That OH^- , NH_4^+ and CN^- radicals are considered as single element when present in a compound therefore $\text{Ca}(\text{OH})_2$, NaCN , NH_4Cl etc. are binary compounds.

Rules in Naming Binary Compounds

Rule 1: The first element, which may be a metal or a non-metal, is named first. Where the first element is NH_4^+ , it is named first as ammonium.

Rule 2: If the first element has a variable oxidation numbers as in $\text{Cu} = +1$ and $+2$, $\text{Fe} = +2$ and $+3$, $\text{Hg} = +1$ and $+2$, $\text{Au} = +1$ and $+3$, $\text{Mn} = +2$, $+4$ and $+7$, $\text{Cr} = +3$ and $+6$, $\text{N} = +1$ to $+5$, $\text{S} = +2$, $+4$ and $+6$, the oxidation number of the element is written in roman figure and is indicated in bracket after the name of the first element e.g. Iron (ii), Sulphur (vi), Nitrogen (iii), Copper (I), Tin (iv).

Rule 3: The second element which is usually a non-metal is named by replacing the last 2, 3 or 4 letters of their name by letter -ide. E.g.

- | | |
|---------------------------|------------------------|
| * Carbon as Carbide | * Chlorine as Chloride |
| * Sulphur as Sulphide | * Oxygen as Oxide |
| * Hydrogen as Hydride | * Fluorine as Fluoride |
| * Phosphorus as Phosphide | * Bromine as Bromide |

Where the second element is OH^- or CN^- radical, it is named as hydroxide or cyanide.

Examples of I.U.P.A.C. Naming of Binary Compounds

Formula	Name
Ca(OH)_2	Calcium hydroxide
SO_3	Sulphur (vi) oxide
SO_2	Sulphur (iv) oxide
Cu_2O	Copper (i) oxide
NaBr	Sodium bromide
CuO	Copper (ii) oxide
FeCl_3	Iron (iii) chloride
NH_4Cl	Ammonium Chloride
NH_4OH	Ammonium hydroxide
Fe_3O_4	Iron (ii) diiron (iii) oxide
NaCN	Sodium cyanide
CaH_2	Calcium hydride
Mg_3N_2	Magnesium nitride

RADICALS

Radicals (polyatomic ions) are group of elements that has electrical charge. They exist only in solution and are not capable of existing alone. Examples are:

Formula	Name
---------	------

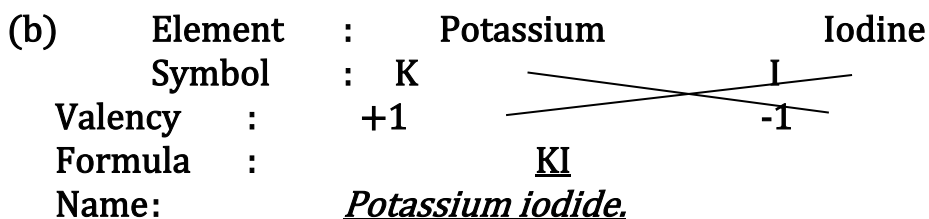
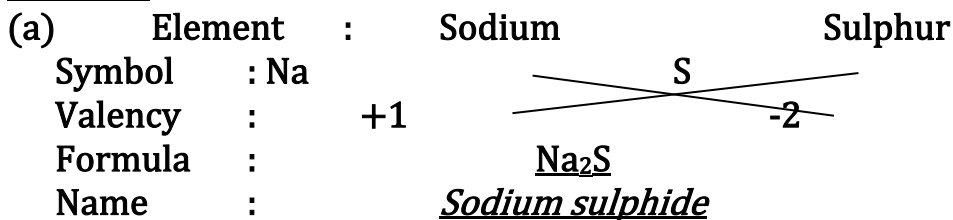
NH_4^+	Ammonium ion
OH^-	Hydroxide ion
CO_3^{2-}	Trioxocarbonate (iv) ion
HCO_3^-	Hydrogen trioxocarbonate (iv) ion
PO_4^{3-}	Tetraoxophosphate (v) ion
HPO_4^{2-}	Hydrogen tetraoxophosphate (v) ion
ClO_3^-	Trioxochlorate (v) ion
MnO_4^-	Tetraoxomanganate (vii) ion
NO_3^-	Trioxonitrate (v) ion
NO_2^-	Dioxonitrate (iii) ion
SiO_3^{2-}	Trioxosilicate (iv) ion
ClO^-	Oxochlorate (i) ion
$\text{S}_2\text{O}_3^{2-}$	Trioxothiosulphate (ii) ion
SO_4^{2-}	Tetraoxosulphate (vi) ion
HSO_4^-	Hydrogen tetraoxosulphate (vi) ion
SO_3^{2-}	Trioxosulphate (iv) ion
CrO_4^{2-}	Tetraoxochromate (vi) ion
$\text{Cr}_2\text{O}_7^{2-}$	Heptaoxodichromate (vi) ion

Formulae of inorganic compounds.

Worked Examples: Write the chemical formulae and chemical name (IUPAC nomenclature) of the compound formed when the following combine together:

- Sodium and sulphur
- Potassium and iodine
- Zinc and trioxocarbonate IV ion
- Iron II and oxygen.

Solution:



(c) Element : Zinc trioxocarbonate IV ion
 Symbol : Zn ~~CO₃⁻²~~
 Valency : +2 ~~-2~~
 Formula : ZnCO₃
 Name: *Zinc trioxocarbonate IV*

(d) Element : Iron II Oxygen
 Symbol : Fe ~~O~~
 Valency : +2 ~~-2~~
 Formula : Fe₂O₂ = FeO
 Name : *Iron II oxide.*

STUDY QUESTIONS

Explain the following terms

1. Chemical symbols
2. Chemical formula
3. Organic compounds
4. Inorganic compounds
5. Radicals
6. Valency
7. Oxidation Number
8. What is the full meaning of I.U.P.A.C.
9. Highlight the four ways in which symbols are assigned to elements.
10. Write the symbols of the first thirty elements.

WEEK 4: EMPIRICAL AND MOLECULAR FORMULAE

OBJECTIVES

At the end of the lesson, students should be able to;

- Explain the meaning of empirical formula and molecular formula
- Solve problems involving empirical and molecular formulae
- Convert empirical formula to molecular formulae and vice versa.

The **empirical formula** is the simplest formula for a compound. The empirical formula is the ratio of elements present in the compound.

Empirical Formulas

Definition

◆ A formula that gives the simplest whole number ratio of the different atoms in a compound.

Ex:

H_2O_2 = hydrogen peroxide
 HO = empirical formula for hydrogen peroxide
 $\text{C}_6\text{H}_{12}\text{O}_6$ = Glucose
Empirical Formula? =

Finding the empirical formula

Example: What is the empirical formula of H_2O_2 ?

Answer: Divide everything by the smallest number in the formula – in this case 2. Dividing through by 2 gives HO .

TASK: Find the empirical formulas of the following:

1. C_6H_6 CH
2. $\text{C}_6\text{H}_{12}\text{O}_6$ CH_2O
3. C_4H_8 CH_2
4. H_2O H_2O
5. $\text{H}_4\text{C}_4\text{O}_8$ $\text{H}_2\text{C}_2\text{O}_4$
6. P_4O_{10} P_2O_5

7. $\text{Ca}(\text{OH})_2$ CaO_2H_2
8. $(\text{NH}_4)_2\text{CO}_3$ $\text{N}_2\text{H}_8\text{CO}_3$
9. $\text{Mg}(\text{NO}_3)_2$ MgN_2O_6

A molecular formula is the same as or a multiple of the empirical formula, and is based on the actual number of atoms of each type in the compound.

The **molecular formula** is the number and type of atoms that are present in a single molecule of a substance.

For example, if the empirical formula of a compound is C_3H_8 , its molecular formula may be C_3H_8 , C_6H_{16} , etc.

Formulas

Empirical formula: the lowest whole number ratio of atoms in a compound.

Molecular formula: the true number of atoms of each element in the formula of a compound.

- ❑ molecular formula = (empirical formula)_n
- ❑ molecular formula = $C_6H_6 = (CH)_6$
- ❑ empirical formula = CH



Empirical vs Molecular Formula



Empirical

Simplest whole number ratio of elements



Molecular

Actual whole number ratio
Multiple of Empirical



sciencenotes.org

Here are the four formulas being used as examples:

Molecular Formula	Empirical Formula
H_2O	H_2O
$\text{C}_2\text{H}_4\text{O}_2$	CH_2O
CH_2O	CH_2O
$\text{C}_6\text{H}_{12}\text{O}_6$	CH_2O

Notice two things:

1. The molecular formula and the empirical formula can be identical.
2. You scale up from the empirical formula to the molecular formula by a whole number factor.

More examples include;

Compound	Molecular formula	Empirical formula
Butane	C_4H_{10}	C_2H_5
Glucose	$\text{C}_6\text{H}_{12}\text{O}_6$	CH_2O
Benzene	C_6H_6	CH
Water	H_2O	H_2O

An empirical formula is often calculated from elemental composition data. The weight percentage of each of the elements present in the compound is given by this elemental composition.

Atomic mass for the first twenty elements;

Element name	Atomic number	Atomic mass	Element name	Atomic number	Atomic mass
Hydrogen	1	1	Sodium	11	23
Helium	2	4	Magnesium	12	24
Lithium	3	7	Aluminium	13	27
Beryllium	4	9	Silicon	14	28
Boron	5	11	Phosphorus	15	31
Carbon	6	12	Sulphur	16	32
Nitrogen	7	14	Chlorine	17	35.5
Oxygen	8	16	Argon	18	40
Fluorine	9	19	Potassium	19	39
Neon	10	20	Calcium	20	40

Empirical Formula: shows the simplest whole number ratio of the atoms present, e.g. C_2H_6 ; C:H ratio is 1:3. and could be written simply as CH_3

Molecular formula is the formula indicating the actual number of atoms of element present in a compound. Being given the molecular mass of a molecule or a compound, the molecular formula could be calculated using its empirical formula

STUDY QUESTIONS

1. What are empirical formulae?
2. Explain molecular formulae
3. Define chemical formulae

Give the empirical formula for the following compounds

4. $\text{C}_6\text{H}_{12}\text{O}_6$
5. H_2O_2
6. H_2O
7. C_6H_6
8. C_4H_{10}
9. P_4O_{10}
10. $\text{C}_{10}\text{H}_{22}$

WEEK 5: LAWS OF CHEMICAL COMBINATIONS

OBJECTIVES:

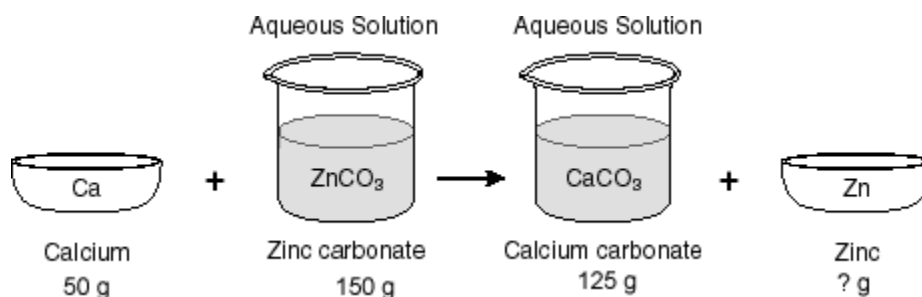
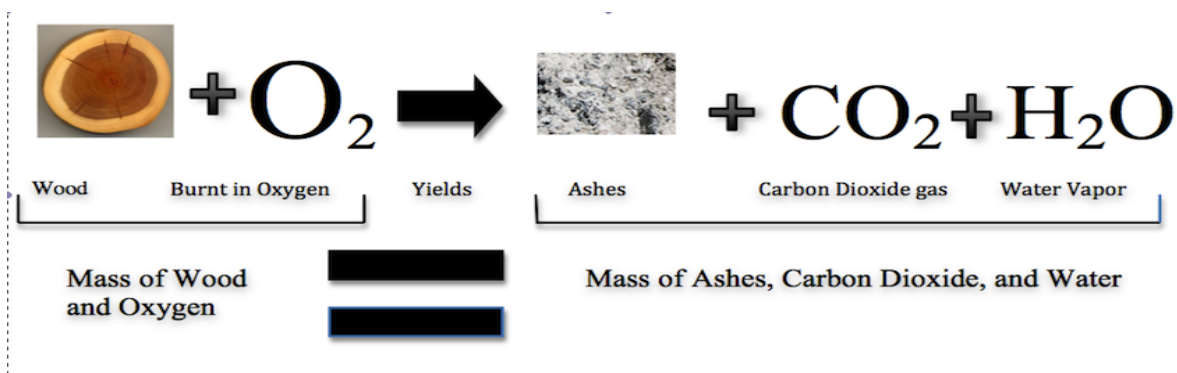
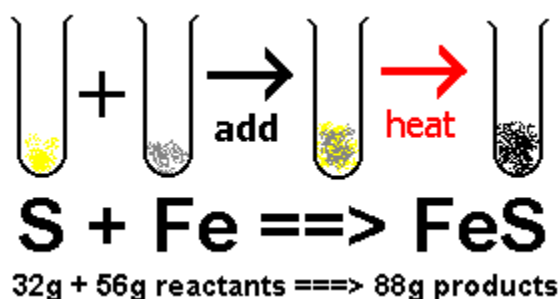
By the end of the lesson, students should be able to;

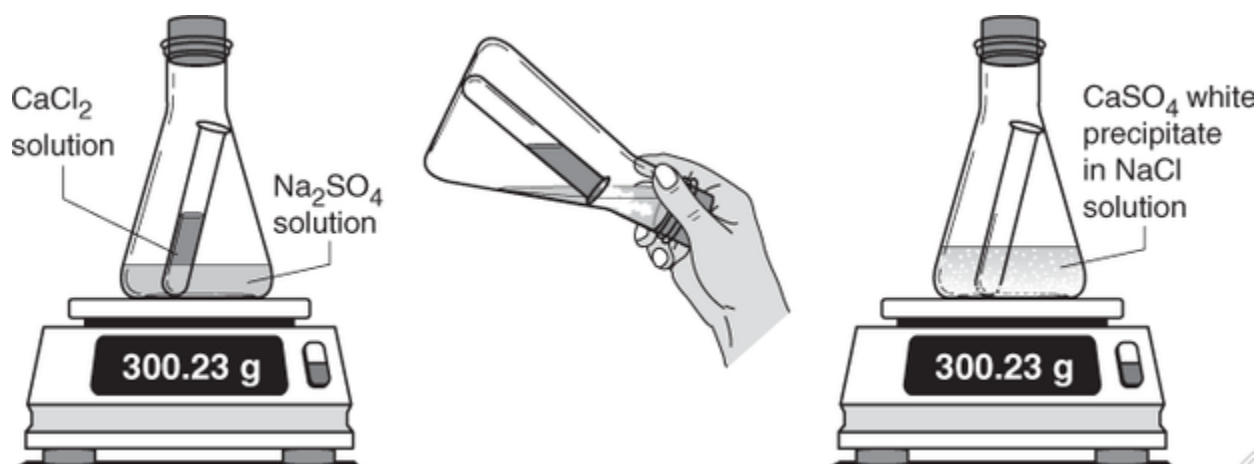
1. State the law of conservation of mass.
2. Describe an experiment to illustrate the law.
3. Perform simple calculations based on the laws

LAWS OF CHEMICAL COMBINATIONS

There are three chemical laws that were put forward by some notable scientist to verify some of the Dalton atomic theory.

1. **Law of Conservation of Matter:** It states that matter can neither be created nor destroyed during chemical reactions.





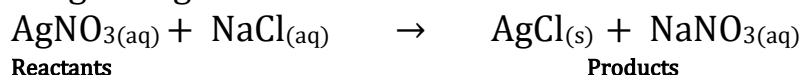
Experiment to verify the law of conservation of matter.

Procedure: Using the precipitation reaction between sodium chloride (NaCl) solution and silver trioxonitrate (v) solution.

Solution of sodium chloride is put inside a conical flask and a solution of silver trioxonitrate (v) is put inside a test tube which is suspended inside the conical flask by means of two strings that are attached to the test tube.

The conical flask is corked using a rubber cork and it is weighed. After the weighing, pull the bottom string of the test tube so that the AgNO_3 solution is poured into the flask for chemical reaction to take place between the $\text{NaCl}_{(\text{aq})}$ and the $\text{AgNO}_{3(\text{aq})}$.

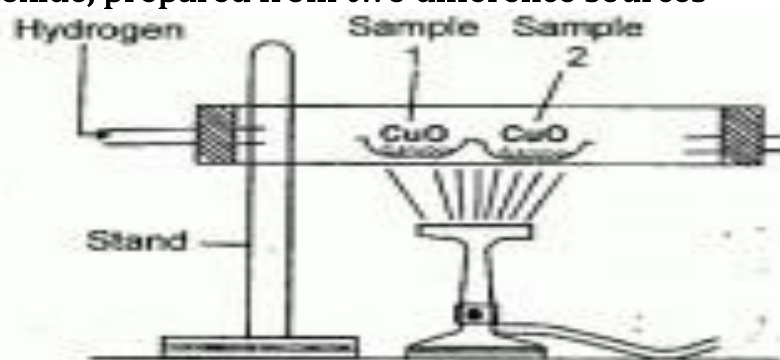
Observation: A white precipitate of silver chloride is formed as a result of chemical reaction between the $\text{NaCl}_{(\text{aq})}$ and the $\text{AgNO}_{3(\text{aq})}$. The flask is then weighed again.



Results: It is found that the mass of the reactants before the chemical reaction is the same as the mass of the silver chloride product at the end of the chemical reaction and thus the law of conservation of matter is verified.

2. **Law of Constant Composition or Definite Proportion:** It states that all pure samples of the same chemical compound contain the same elements combined in the same proportion by mass.

Experiment to determine the proportion of copper in the two samples of pure copper (ii) oxide, prepared from two difference sources



Verification of the law of definite proportion

Procedure: The experiment is set up as in the above diagram. Two different samples of pure samples of copper (II) oxide (CuO) prepared from two different sources (i and ii above) and labelled samples A and sample B respectively are placed in two separate metal boats which are weighed and the weights recorded. The weighed metal boats are placed inside a hard glass combustion tube with a jet flame at its one end.

Excess Hydrogen gas used for the experiment burnt off at the jet flame. The hard glass combustion tube is clamped to two retort stands in a slanted position in order to prevent water from running back into the hot tube. The tube is heated using two Bunsen burners and stream of Hydrogen gas is passed over the heated CuO in the tube, the Hydrogen gas reduces the black copper (ii) oxide to reddish brown copper.

The hydrogen gas is continuously passed over the hot copper until the tube is completely cooled in order to prevent oxidation of the hot copper back to copper (ii) oxide.

The two metal boats containing the copper are weighed and the weight recorded. Below are the results of the experiment.

	Oxide A	Oxide B
Mass of CuO	10.59	3.18
Mass of Cu	5.00	1.50
Mass of oxygen	5.59	1.68

Question: In an experiment, 10.59g of copper (ii) oxide is reduced to 5.00g of copper on reduction in a stream of hydrogen gas. In the same experiment, 3.18g of copper (ii) oxide is reduced to 1.50g of copper. Show that the results are in accordance with the law of constant composition.

3. Law of Multiple Proportions

It states that if two element A and B combine to form more than one compound, the different masses of element A which combine with fixed mass of element B are in simple whole number ratio.

Practical Experiment To Verify The Law of Multiple Proportions

The same experiment for the law of constant composition is carried out but two different samples of copper oxide i.e copper (ii) oxide (CuO) and copper (i) oxide (Cu₂O) are used for the experiment and the samples are labelled sample A and sample B respectively.

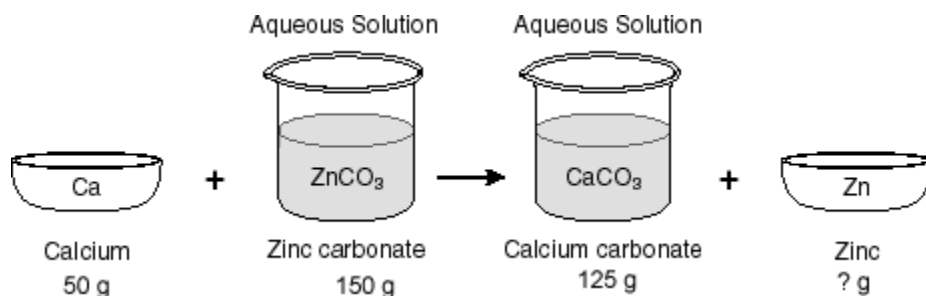
The following results are obtained for the experiment on law of multiple proportions.

	Oxide A	Oxide B
Mass of Oxide	2.85g	1.90g
Mass of Copper	2.50g	1.50g
Mass of oxygen	0.35g	0.40g

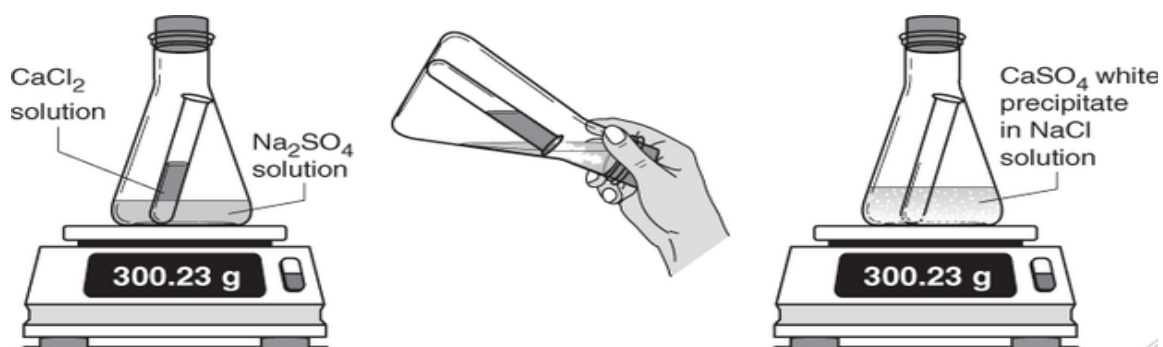
STUDY QUESTIONS

State the following laws

1. law of conservation of matter.
2. Law of Constant Composition or Definite Proportion
3. Law of Multiple Proportions
4. Another name given to the law of conservation of matter is?
5. State the second law of thermodynamics
6. Briefly describe an experiment to illustrate the law of conservation of matter.
7. From the illustration below, give the mass of zinc

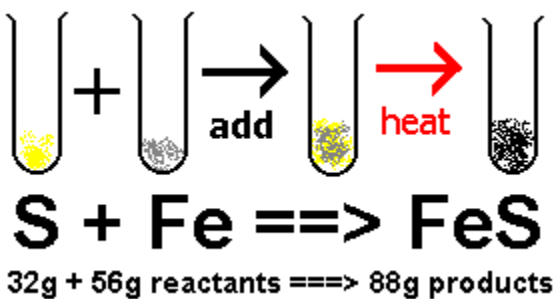


8. Explain the reason for the equal mass of both solutions on the weighing scale.



9. Highlight (2) Dalton atomic theory

10. The equation below shows a chemical reaction and is represented in chemical symbols, write it in a word equation form.



WEEK 6: CHEMICAL EQUATIONS

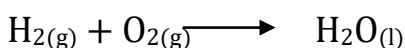
OBJECTIVES:

- i. Write the equation for the reaction between HCl and CaCO₃ to form water, CO₂ and CaCl₂.
- ii. Balance simple chemical equations.

A chemical reaction takes place when different elements combine with each other to form a new substance. A chemical equation therefore is a condensed or shorthand statement about a chemical reaction. It represents the changes in bonding and energy that takes place during a chemical reaction.

A chemical equation has two parts:

- i. The right-hand side i.e., the product side
- ii. The left-hand side i.e., the reactant side.



reactants product

The plus sign means “reacting with” while the arrow sign means “produce” or “yield” or “give”.

A chemical equation describes what happens in a chemical reaction. The equation identifies the reactants (starting materials) and products (resulting substance), the formulas of the participants, the phases of the participants (solid, liquid, gas), and the amount of each substance.

BALANCING CHEMICAL EQUATIONS

Balancing a chemical equation refers to establishing the mathematical relationship between the quantity of reactants and products. The quantities are expressed as grams or moles.

A chemical equation is a condensed statement of a chemical reaction. It represents the changes in bonding and energy that takes place in a reaction.

According to the **law of conservation of matter/mass**, atoms are neither created nor destroyed in a chemical reaction, they are only changed from one form to another. So, the total number of an atom present in the reactant side must be equal to the total number of the same atom present on the product side.

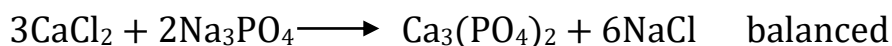
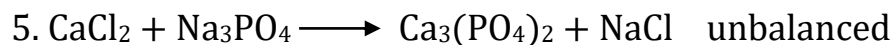
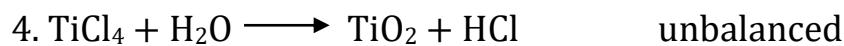
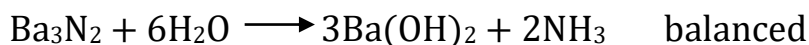
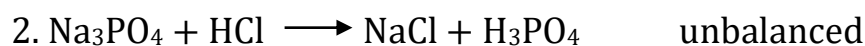
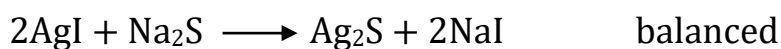
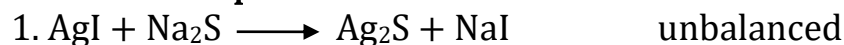
We use coefficients to indicate the smallest number of molecules of a substance that can take part in a chemical reaction. **COEFFICIENTS** are numbers written in front of formulae to balance the equation.

A balanced equation accounts for all the atoms taking part in a chemical reaction both in number and in kind. It takes practice to be able to write balanced equations.

When balancing chemical equations, the following should be noted.

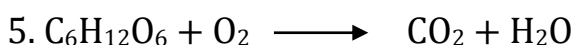
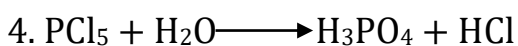
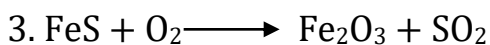
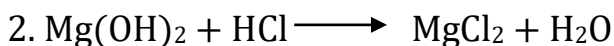
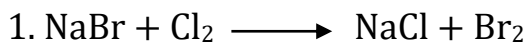
1. Equations must be balanced by using coefficients in front of the formulae and not by changing the subscript numbers within the formulae.
2. Common gases such as hydrogen, oxygen, nitrogen, chlorine etc. in the free state are diatomic e.g. H_2 , O_2 , N_2 and Cl_2 .
3. Metals in their free state are represented by their atomic symbols e.g. Cu, Fe, Na, K, Mg etc.
4. In simple chemical reactions, radicals remain unchanged during the reactions.

Worked examples:

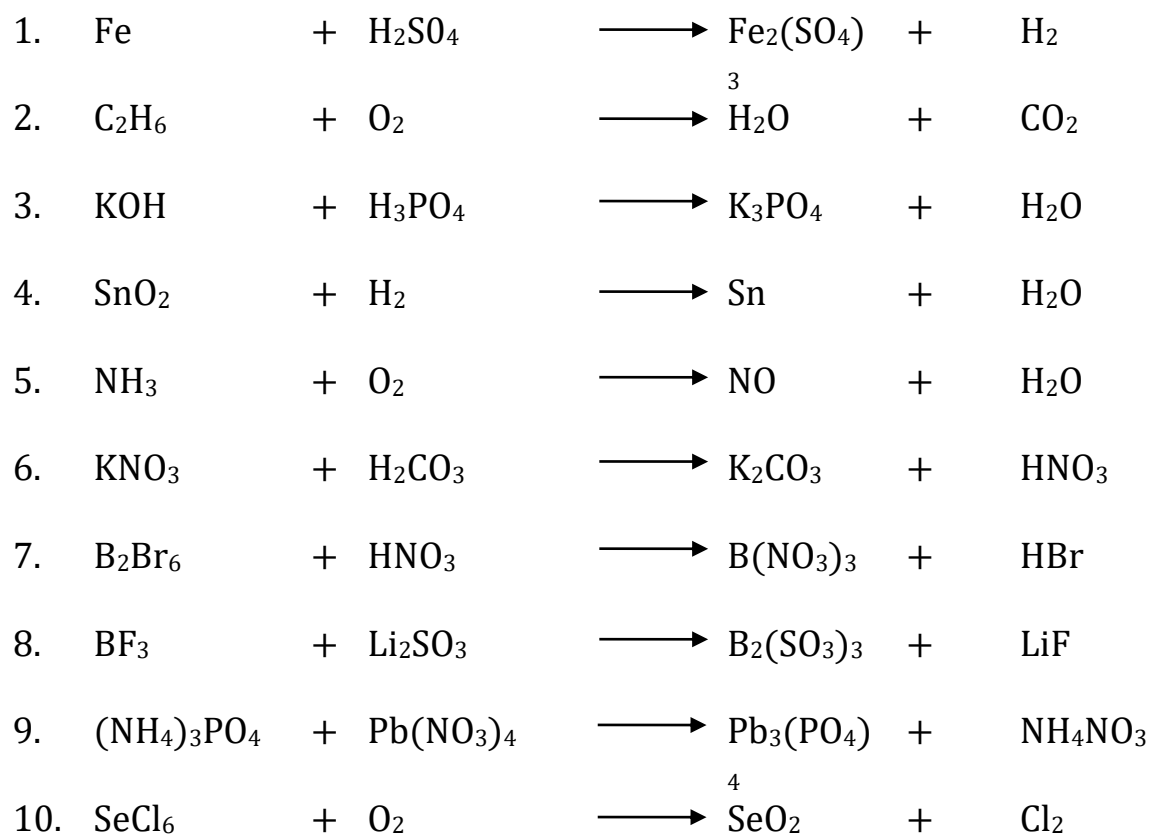


CLASSWORK

Balance the following chemical equations



STUDY QUESTIONS:



WEEK 8

CHEMICAL REACTIONS

OBJECTIVES: By the end of the lesson, students should be able to;

- Differentiate between exothermic and endothermic reactions.
- Give THREE examples EACH of exothermic and endothermic reactions.

A chemical reaction is a process that is usually characterized by a chemical change in which the starting materials (reactants) are different from the products. Chemical reactions tend to involve the motion of electrons, leading to the formation and breaking of chemical bonds. There are several different types of chemical reactions and more than one way of classifying them.

Exothermic and Endothermic Reactions:

During a chemical reaction, heat can be evolved or absorbed depending on the relative heat content of the reactants and the products. When the total heat content of the products is less than that of the reactants, the excess energy will be given off as heat to the surrounding as the reaction occurs. This type of reaction is said to be **exothermic**.

When the total heat content of the product is more than that of the reactants, heat energy must be absorbed from the surroundings for the reaction to proceed. Such a reaction is said to be **endothermic**.

An **exothermic** reaction is one during which heat is liberated to the surroundings.

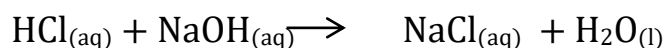
An **endothermic** reaction is one during which heat is absorbed from the surroundings.

Some examples of exothermic reactions are as follows:

- Reaction between calcium oxide and water

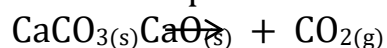


- Reaction between an acid and a base



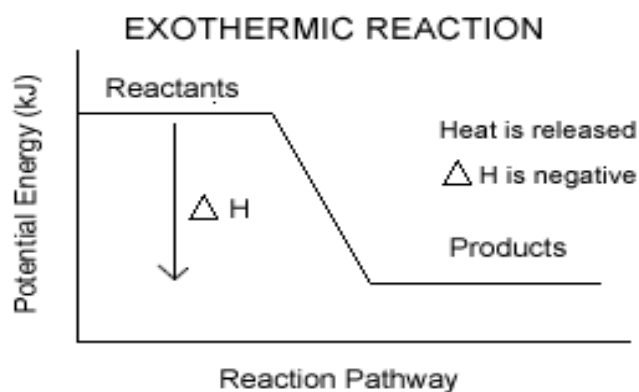
Some examples of endothermic reactions are as follows:

- Thermal decomposition of calcium trioxocarbonate (IV)

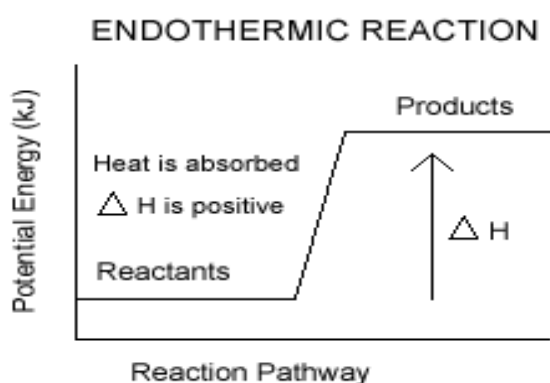


- Thermal dissociation of ammonium chloride





Endothermic



STUDY QUESTIONS

What do you understand by;

1. Exothermic and
2. Endothermic reactions
3. Differentiate between exothermic and endothermic reactions.
4. Give THREE examples EACH of exothermic reactions.
5. Give THREE examples EACH of endothermic reactions.
6. Illustrate (1) chemical equation for exothermic reaction
7. Illustrate (1) chemical equation for endothermic reaction
8. What is a chemical reaction
9. Chemical reactions, results in _____ and _____ of chemical bonds.
10. The starting materials in a chemical reaction are known as _____

WEEKS9 AND 10: TYPES OF REACTIONS

OBJECTIVES

- i. List SIX types of chemical reactions
- ii. Define combination reaction.
- iii. Differentiate between the following pairs:
 - (a) Thermal decomposition reaction and thermal dissociation.
 - (b) Homogeneous catalysis and heterogeneous catalysis

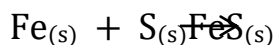
Chemical reactions are changes in which new chemical substances are formed. Chemical reactions can be grouped as:

- Combination
- Decomposition
- Displacement
- Double decomposition
- Catalytic reaction
- Reversible reactions
- Thermal dissociation
- Oxidation and reduction

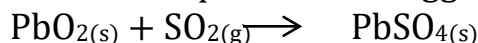
1. **Combination reaction:** This is a reaction in which two or more substances combine to form a single compound.

Combinations can occur in any one of the following ways:

Element + element $\longrightarrow$ compound e.g



Compound + compound $\longrightarrow$ bigger compound e.g.

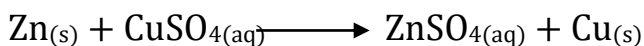


Many organic compounds also undergo combination reactions. The polymerization of ethyne, for example, is a special type of combination reaction.

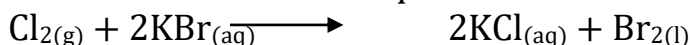
2. **Decomposition reaction:** This is a reaction in which a single compound split up into two or more simpler substances. When heat is needed to bring about decomposition, the reaction is called **thermal decomposition**.



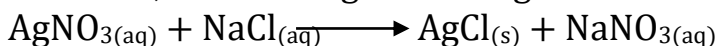
3. **Displacement reaction:** This is a reaction in which an element (or a radical) replaces another element (or a radical) in a compound. The ability of an element to displace another is determined by their relative positions in the electrochemical series. Thus more electropositive metals will displace less electropositive metals from their aqueous solutions. E.g.



Also, more electronegative non-metal will displace less electronegative non-metals from their aqueous solutions. E.g.



4. **Double decomposition:** This is a reaction in which the reactants decompose to form new substances by an exchange of radicals. The two reactants are soluble while one of the products is soluble while the other is insoluble, volatile or gaseous. E.g.

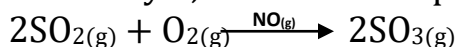


5. **Catalytic reaction:** This is a chemical reaction during which a catalyst is used. The acceleration or retardation of reactions by catalysts is known as **catalysis**.

A catalyst is a substance which alters the rate of a chemical reaction, but which itself remains chemically and quantitatively unchanged at the end of the reaction.

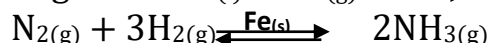
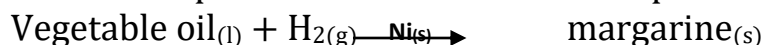
There are two types of catalysed reactions and these are:

- i. **Homogeneous catalysis:** This is a catalysed reaction in which the catalyst, reactants and products are all in the same phase. E.g.



- ii. **Heterogeneous catalysis:** This is a catalysed reaction in which the catalyst,

reactants and products are in the different phase. E.g.



NOTE: Since catalysts do not affect the overall reaction, they are written above the arrow in an equation.

Characteristics of a catalyst

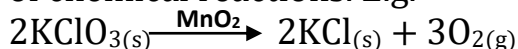
- i. A catalyst alters the rate of a chemical reaction.
- ii. It remains unchanged in chemical nature and in mass at the end of a reaction. Its physical features like colour and texture may change.
- iii. A catalyst is specific in action i.e. a given catalyst will act on only one particular reaction. Many inorganic catalysts can catalyse more than one reaction.
- iv. A catalyst cannot start a reaction. It only affects a reaction which is already in progress.
- v. A catalyst does not affect the type of product formed in a reaction.
- vi. A catalyst has no effect on the equilibrium of a reversible reaction.
- vii. A catalyst will affect the rate of a reaction even though it is present in very small amounts. An increase in the amount of catalyst up to a certain limit would usually increase the reaction rate.

viii. The effect of a solid catalyst is improved by increasing its surface area.

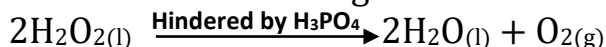
Types of catalysts

a. **Inorganic catalysts:** These are catalysts that are mineral elements or compounds. They can be grouped into:

i. **Positive catalysts:** These are catalysts which increases the rate of chemical reactions. E.g.



ii. **Negative catalysts:** These are catalysts which decreases the rate of chemical reactions. E.g.



b. **Organic catalysts:** These are known as enzymes. Enzymes are organic catalysts which control the rate of biochemical reactions in living organisms. They are produced by living cells. They help to digest food, cause blood to clot, make palm wine and fruit juices ferment and are also used in the industrial manufacture of alcohol from sugars and starches.

Promoters and inhibitors: These are not catalysts but they influence the rate of a reaction by affecting the efficiency of the catalysts.

Promoters are substances which improve the efficiency of a catalyst.

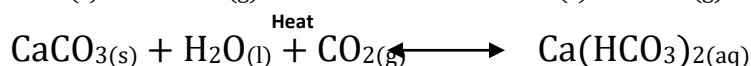
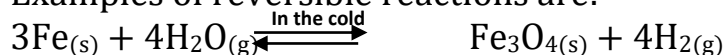


The iron catalyst used in Haber process is activated by the addition of trace amounts of aluminium oxide or molybdenum.

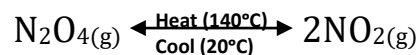
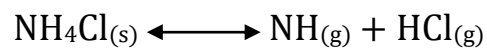
Inhibitors are substances which reduces the efficiency of a catalyst. They are also known as catalyst poisons. Hydrogen sulphide, arsenic (III) oxide, hydrogen cyanide and mercury salts are catalyst poisons which harm us by inhibiting important enzymes in the body.

6. **Reversible reactions:** A reversible reaction is a chemical reaction where the reactants form products that in turn react together/split to give the reactants back. Reversible reactions will reach an equilibrium point where the concentrations of the reactants and products will no longer change.

A reversible reaction is denoted by a double arrow pointing both directions in a chemical equation. The conditions for the reverse reaction may or may not be the same as the conditions for the forward reaction. Examples of reversible reactions are:



7. **Thermal dissociation:** This is a chemical reaction in which a reactant molecule separates into two or more simpler molecules or atoms on the application of heat. Unlike thermal decomposition, thermal dissociation is reversible. Examples are:



STUDY QUESTIONS

Explain the following reactions

1. Combination
2. Decomposition
3. Displacement
4. Double decomposition
5. Catalytic reaction
6. Reversible reactions
7. Thermal dissociation
8. Oxidation and reduction

Differentiate between the following pairs:

9. Thermal decomposition reaction and thermal dissociation.
10. Homogeneous catalysis and heterogeneous catalysis